

Linear Algebra

Personal Notes

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1 The Geometry of Linear Equations

The basic computational problem of linear algebra is to solve a *system of linear equations*. A system with n equations and n unknowns can be understood in three complementary ways: one equation at a time, one column at a time, or as a single matrix equation.

1.1 A two-dimensional system

Consider

$$\begin{aligned}2x - y &= 0, \\ -x + 2y &= 3.\end{aligned}$$

In the *row picture*, each equation describes a line in the xy -plane. The first line passes through the origin and the point $(1, 2)$; the second contains $(-3, 0)$ and $(-1, 1)$. Their intersection is $(1, 2)$, so the simultaneous solution is

$$x = 1, \quad y = 2.$$

Substitution checks both equations. This picture makes the meaning of a solution immediate: it is a point satisfying every row at once.

The same system has a different geometric interpretation in the *column picture*. Grouping coefficients by unknown gives

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

A *linear combination* of vectors $c_1, \dots, c_k$ is an expression $a_1c_1 + \dots + a_kc_k$ with scalar coefficients a_i . Solving the system therefore means choosing the combination of the two coefficient columns that produces the right-hand side. Here

$$1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix},$$

which again yields $x = 1$ and $y = 2$.

1.2 The matrix equation

The compact algebraic form is

$$\underbrace{\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 0 \\ 3 \end{bmatrix}}_{\mathbf{b}}, \quad A\mathbf{x} = \mathbf{b}.$$

The matrix A is the *coefficient matrix*, $\mathbf{x}$ is the vector of unknowns, and $\mathbf{b}$ is the right-hand-side vector.

The *matrix–vector product* $A\mathbf{x}$ should first be viewed as a combination of the columns of A . If

$$A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

then

$$A\mathbf{x} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}.$$

Equivalently, each component is the dot product of a row of A with $\mathbf{x}$:

$$A\mathbf{x} = \begin{bmatrix} 2 \cdot 1 + 5 \cdot 2 \\ 1 \cdot 1 + 3 \cdot 2 \end{bmatrix}.$$

The row calculation and column combination are the same multiplication seen from different directions.

1.3 Three equations and three unknowns

For the system

$$\begin{aligned} 2x - y &= 0, \\ -x + 2y - z &= -1, \\ -3y + 4z &= 4, \end{aligned}$$

the matrix form is

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}.$$

In the row picture, each equation is a plane in three-dimensional space. Two planes generally meet along a line, and a third plane can reduce that common set to one point. That point is the solution. The idea is correct but the geometry quickly becomes difficult to draw in higher dimensions.

The column picture remains algebraically useful. It asks for scalars x, y, z such that

$$x \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}.$$

Because the right-hand side is the third column, the solution is immediately $\mathbf{x} = (0, 0, 1)^T$. A different right-hand side changes the required coefficients but not the columns being combined.

1.4 Which right-hand sides are reachable?

The equation $A\mathbf{x} = \mathbf{b}$ is solvable precisely when $\mathbf{b}$ is a linear combination of the columns of A . Thus the question

“Can we solve $A\mathbf{x} = \mathbf{b}$ for every $\mathbf{b}$?”

is equivalent to asking whether the column combinations fill the entire ambient space.

An n -by- n matrix is *invertible* (or *nonsingular*) when every right-hand side has exactly one solution. Geometrically, its columns provide enough independent directions to reach any vector. By contrast, columns are *linearly dependent* when one can be formed from the others. In three dimensions, if all three columns lie in one plane, every combination remains in that plane. Right-hand sides outside the plane are unreachable, and the matrix is *singular*.

This column viewpoint scales beyond what can be drawn: even for nine equations and nine unknowns, $A\mathbf{x}$ is still a linear combination of nine columns. The next task is to develop elimination, a systematic procedure for finding the coefficients when a solution exists.

Source: MIT OpenCourseWare, 18.06SC Linear Algebra (Fall 2011), Session 1, [The Geometry of Linear Equations](#), lecture by Gilbert Strang.

2 Elimination with Matrices

Elimination solves a linear system by replacing equations with equivalent ones until the coefficient matrix has a form that is easy to solve. For an invertible matrix, the procedure produces one pivot for each unknown and then recovers the solution in reverse order.

2.1 Forward elimination

Consider

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 12 \\ 2 \end{bmatrix}, \quad A\mathbf{x} = \mathbf{b}.$$

The entry 1 in position (1, 1) is the first *pivot*. A pivot is the nonzero entry used to eliminate the entries below it. The corresponding *multiplier* for row 2 is $3/1 = 3$, so the first row operation is

$$R_2 \leftarrow R_2 - 3R_1.$$

Row 3 already has a zero beneath the first pivot. The matrix becomes

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}.$$

The second pivot is 2. Its multiplier for row 3 is $4/2 = 2$, giving

$$R_3 \leftarrow R_3 - 2R_2$$

and hence

$$A \longrightarrow U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}.$$

The matrix U is *upper triangular*: every entry below its main diagonal is zero. The pivots are 1, 2, 5.

Every row operation must also be applied to the right-hand side. Appending $\mathbf{b}$ to A creates the *augmented matrix*

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ 0 & 4 & 1 & 2 \end{array} \right].$$

The same two operations transform it into

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 5 & -10 \end{array} \right],$$

so $A\mathbf{x} = \mathbf{b}$ has become $U\mathbf{x} = \mathbf{c}$, with $\mathbf{c} = (2, 6, -10)^T$.

2.2 Back substitution and pivot failure

The triangular system is

$$\begin{aligned} x + 2y + z &= 2, \\ 2y - 2z &= 6, \\ 5z &= -10. \end{aligned}$$

Back substitution starts with the last equation and moves upward:

$$z = -2, \quad y = 1, \quad x = 2.$$

Thus $\mathbf{x} = (2, 1, -2)^T$, which also solves the original system.

The complete reduction from A to U is called *forward elimination*. A zero cannot serve as a pivot. If the pivot position contains zero but a lower entry in the same column is nonzero, a row exchange moves that entry into the pivot position. If the pivot and every entry below it are zero, no exchange can repair the step. The square matrix is then singular: some right-hand sides have no solution, while others may have infinitely many, so a unique solution for every right-hand side is impossible.

2.3 Elimination as matrix multiplication

Row operations can themselves be represented by matrices. The *elimination matrix*

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

performs $R_2 \leftarrow R_2 - 3R_1$ when it multiplies on the left:

$$E_{21}A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}.$$

The unchanged rows of E_{21} are rows of the *identity matrix*

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

which satisfies $IA = A$.

The next operation is encoded by

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}.$$

Consequently,

$$E_{32}(E_{21}A) = U.$$

Matrix multiplication is *associative*, so parentheses may move without changing the order of the factors:

$$E_{32}(E_{21}A) = (E_{32}E_{21})A.$$

The product $E = E_{32}E_{21}$ performs both elimination steps. Applying the same matrix to both sides gives

$$E(A\mathbf{x}) = E\mathbf{b} \iff (EA)\mathbf{x} = E\mathbf{b} \iff U\mathbf{x} = \mathbf{c}.$$

The side of multiplication determines what is changed. Left multiplication combines rows, while right multiplication combines columns. Matrix multiplication is generally not commutative: PA and AP need not agree.

2.4 Row exchanges and inverses

A *permutation matrix* is obtained by rearranging rows of I . For example,

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

exchanges the first two rows of A in PA . Multiplying as AP instead exchanges the first two columns.

An *inverse* of a matrix E is a matrix E^{-1} satisfying $E^{-1}E = I$. Since E_{21} subtracts three times row 1 from row 2, its inverse adds that quantity back:

$$E_{21}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_{21}^{-1}E_{21} = I.$$

Each successful elimination step is reversible. This is why the transformed system has the same solution as the original one, and it also explains how a sequence of inverse elimination matrices can reconstruct A from U .

Source: MIT OpenCourseWare, 18.06SC Linear Algebra (Fall 2011), Session 2, [Elimination with Matrices](#), lecture by Gilbert Strang.

3 Multiplication and Inverse Matrices

Matrix multiplication can be understood entry by entry, one column at a time, one row at a time, as a sum of column–row products, or in blocks. Each view exposes a different part of the same calculation.

3.1 The row–column rule

Let A be m -by- n and B be n -by- p . Their *matrix product* $C = AB$ is the m -by- p matrix whose (i, j) entry is

$$c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}.$$

Thus the number of columns of A must equal the number of rows of B ; the two outer dimensions determine the shape of the answer.

This *row–column rule* says that c_{ij} is the dot product of row i of A with column j of B . For example,

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 4 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} 1 \cdot 2 + 2 \cdot 0 + 0 \cdot 5 & 1 \cdot 1 + 2 \cdot 4 + 0 \cdot (-2) \\ -1 \cdot 2 + 3 \cdot 0 + 1 \cdot 5 & -1 \cdot 1 + 3 \cdot 4 + 1 \cdot (-2) \end{bmatrix} = \begin{bmatrix} 2 & 9 \\ 3 & 9 \end{bmatrix}.$$

3.2 Columns, rows, and outer products

Write the columns of B as $\mathbf{b}_1, \dots, \mathbf{b}_p$. Then

$$AB = [A\mathbf{b}_1 \quad \cdots \quad A\mathbf{b}_p].$$

Every column of AB is therefore a linear combination of the columns of A , with coefficients supplied by the corresponding column of B .

The same product can be read by rows. If $\mathbf{a}_i^\top$ is row i of A , then row i of AB is

$$\mathbf{a}_i^\top B.$$

Consequently, every row of AB is a linear combination of the rows of B . The column view explains where the output columns come from; the row view explains the output rows.

Multiplying a column $\mathbf{u}$ by a row $\mathbf{v}^\top$ produces the *outer product* $\mathbf{u}\mathbf{v}^\top$. For instance,

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 3 & 18 \\ 4 & 24 \end{bmatrix}.$$

All columns of this matrix are multiples of $(2, 3, 4)^\top$, and all rows are multiples of $(1, 6)$.

If $\mathbf{a}_k$ is column k of A and $\mathbf{r}_k^\top$ is row k of B , then

$$AB = \sum_{k=1}^n \mathbf{a}_k \mathbf{r}_k^\top.$$

Matrix multiplication is therefore a sum of outer products, one for each matched inner index.

3.3 Block multiplication

A *block matrix* is a matrix partitioned into smaller matrices. Blocks may be multiplied by the same row-column rule whenever their dimensions are compatible. For example,

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} = \begin{bmatrix} A_1 B_1 + A_2 B_3 & A_1 B_2 + A_2 B_4 \\ A_3 B_1 + A_4 B_3 & A_3 B_2 + A_4 B_4 \end{bmatrix}.$$

The blocks need not all be square or equally sized; only the partitions along each multiplication and addition must match.

3.4 When does an inverse exist?

For a square matrix A , an inverse satisfies

$$A^{-1}A = I = AA^{-1}.$$

The existence of A^{-1} is equivalent to being able to solve $A\mathbf{x} = \mathbf{b}$ uniquely for every right-hand side. Indeed, if $\mathbf{e}_j$ is column j of the identity and $\mathbf{x}_j$ is column j of A^{-1} , then

$$A\mathbf{x}_j = \mathbf{e}_j.$$

Finding the inverse is therefore the same as solving n systems with the same coefficient matrix and the n columns of I as right-hand sides.

A nonzero solution of $A\mathbf{x} = \mathbf{0}$ proves that A is singular. For example,

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

If an inverse existed, multiplying by it would give $\mathbf{x} = A^{-1}A\mathbf{x} = \mathbf{0}$, contradicting $\mathbf{x} = (3, -1)^\top \neq \mathbf{0}$. Geometrically, the two columns lie on the same line, so their combinations cannot produce every right-hand side.

3.5 Gauss–Jordan elimination

Gauss–Jordan elimination computes an inverse by augmenting A with the entire identity matrix and applying row operations until the left half becomes I :

$$\left[A \mid I \right] \longrightarrow \left[I \mid A^{-1} \right].$$

For

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix},$$

the two elimination steps are

$$\begin{aligned} \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array} \right] &\xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right] \\ &\xrightarrow{R_1 \leftarrow R_1 - 3R_2} \left[\begin{array}{cc|cc} 1 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{array} \right]. \end{aligned}$$

Hence

$$A^{-1} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}.$$

The block calculation explains why the method works. If E is the product of all row-operation matrices, then

$$E \left[A \mid I \right] = \left[EA \mid E \right].$$

Reaching I on the left means $EA = I$, so $E = A^{-1}$; the right half must therefore be the inverse. If elimination cannot reduce the left half to I , then A is singular and no inverse exists.

Source: MIT OpenCourseWare, 18.06SC Linear Algebra (Fall 2011), Session 3, [Multiplication and Inverse Matrices](#), lecture by Gilbert Strang.

4 Factorization into $A = LU$

Gaussian elimination can be recorded as a factorization of the original matrix. Before constructing it, two order-reversing rules explain how the elimination matrices move from one side of an equation to the other.

4.1 Products, inverses, and transposes

If A and B are invertible, then

$$(AB)^{-1} = B^{-1}A^{-1}.$$

The order reverses because adjacent factors must cancel:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = I.$$

The *transpose* A^T is obtained by exchanging the rows and columns of A ; equivalently, $(A^T)_{ij} = A_{ji}$. Transposing a product also reverses its order:

$$(AB)^T = B^T A^T.$$

For an invertible matrix, transposing and taking an inverse commute:

$$(A^T)^{-1} = (A^{-1})^T.$$

Indeed, transposing $A^{-1}A = I$ gives $A^T(A^{-1})^T = I$.

4.2 From elimination to $A = LU$

Suppose elimination reaches an upper triangular matrix U without any row exchanges. If E is the product of the elimination matrices, then

$$EA = U.$$

Multiplying by E^{-1} gives

$$A = E^{-1}U = LU.$$

This is the *LU factorization*. The matrix $L = E^{-1}$ is *lower triangular*: every entry above its diagonal is zero. In this factorization, L has ones on its diagonal and contains the elimination multipliers below it.

For example, eliminating the 8 below the first pivot of

$$A = \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}$$

uses multiplier 4:

$$\underbrace{\begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix}}_{E_{21}} \underbrace{\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}}_U.$$

The inverse operation adds four times row 1 back to row 2, so

$$A = LU = \underbrace{\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}}_U.$$

The multiplier appears in L with a positive sign, while the elimination matrix contains its negative.

4.3 Several elimination steps

For a three-by-three matrix, elimination without row exchanges has the form

$$E_{32}E_{31}E_{21}A = U.$$

The inverse-of-a-product rule reverses the order when the factors move to the right:

$$A = \underbrace{E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}}_L U.$$

The reverse order prevents earlier row operations from creating unwanted cross-terms in L . For example, take $E_{31} = I$ and elimination multipliers 2 and 5. The product of elimination matrices is

$$E_{32}E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 10 & -5 & 1 \end{bmatrix}.$$

The factor used in $A = LU$, however, is

$$L = E_{21}^{-1}E_{32}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}.$$

Thus, when no row exchanges occur, the multipliers can be copied directly into L while the evolving upper triangular entries are retained in U . There is no need to store the individual elimination matrices.

4.4 Separating the pivots

Sometimes it is useful to split the pivots from U . A *diagonal matrix* has zero entries away from its main diagonal. Writing $U = DU'$, where D contains the pivots and U' has ones on its diagonal, gives

$$A = LDU'.$$

For the two-by-two example,

$$\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}.$$

4.5 Cost of elimination

For a dense n -by- n matrix, the successive elimination stages act on smaller and smaller trailing matrices. Their leading operation count is

$$1^2 + 2^2 + \cdots + n^2 \sim \int_0^n x^2 dx = \frac{1}{3}n^3.$$

Thus computing L and U costs about $n^3/3$ combined multiply-subtract steps. Once the factorization is available, carrying one right-hand side through forward and back substitution costs only about n^2 operations. This makes one factorization especially valuable when many systems share the same matrix A .

4.6 Row exchanges

If a zero appears in a pivot position, a row exchange is required. Left multiplication by

$$P_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

swaps rows 1 and 2 of a three-by-three matrix. A permutation matrix is obtained by rearranging the rows of I , so there are $n!$ such matrices of size n -by- n . Their inverses satisfy

$$P^{-1} = P^T.$$

The simple $A = LU$ construction above assumes that no such exchanges are needed; permutation matrices keep track of them when they occur.

Source: MIT OpenCourseWare, 18.06SC Linear Algebra (Fall 2011), Session 4, [Factorization into \$A = LU\$](#) , lecture by Gilbert Strang.